

Flow Matching as a Classification Layer

Stage 1 — frozen-encoder baselines · Stage 2 — an FM layer on top of them

3 × 2

datasets × encoders

DTD, FGVC-Aircraft, Flowers-102
ResNet-18, DINOv2 ViT-S/14

270

velocity networks trained

162 image-prototype branch
108 CLIP text-prototype branch

+24.3

points, best Δ Acc

FGVC-Aircraft / DINOv2, full data
over the Stage 1 prototype baseline

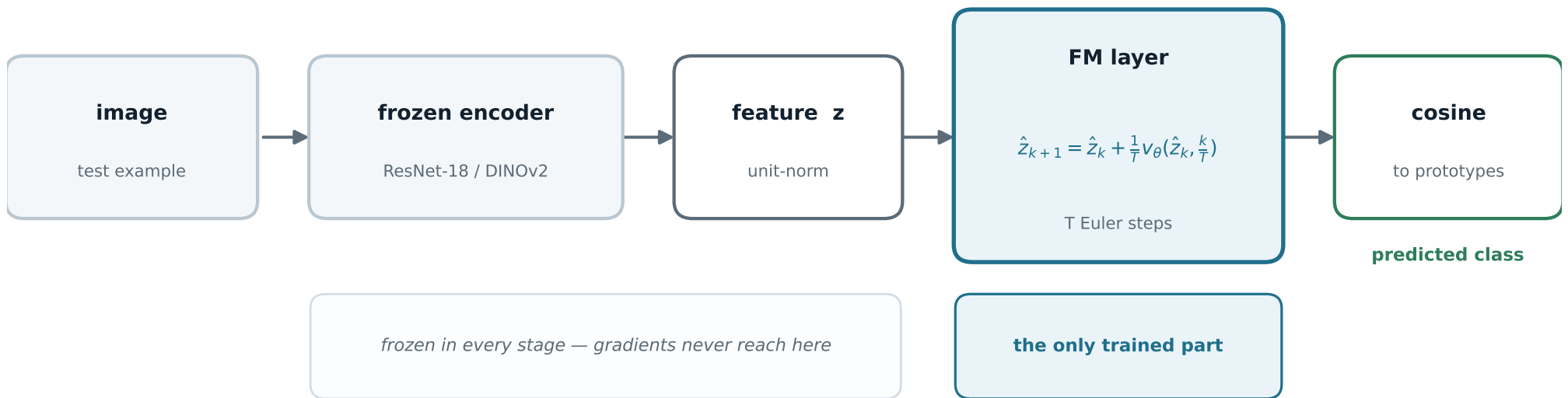
What this report contains

The method and the two training objectives, drawn; the experimental protocol; and every measured result from the executed notebooks. All accuracies are top-1 on the complete official test splits and are read directly from the saved run metrics — no number in this document was typed by hand.

The headline result, stated honestly

The FM layer improves on the prototype baseline it replaces in 66 of 72 result cells, by as much as +24.3 points. It does not beat a linear probe on the same features. And on the CLIP branch, where a fair same-supervision control exists, the FM layer loses to that control in 28 of 36 cells — the large gains against zero-shot there measure the labels, not the layer.

What the Flow Matching layer does



The idea

Stage 1 classifies a frozen feature by cosine similarity to a fixed class prototype. Stage 2 inserts a learned transport in front of that rule: a velocity field carries the feature toward the prototype of its class, before the same cosine comparison is applied. The encoder and the prototypes stay frozen, so any accuracy change is attributable to the layer alone.

The catch worth stating early

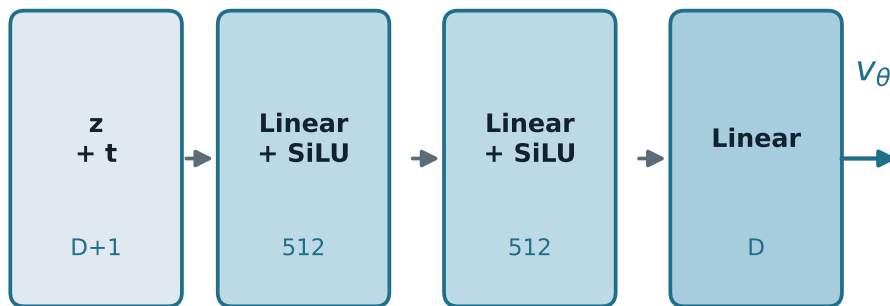
At test time the network cannot know the label, so it cannot learn "move z to p_y ". It learns the conditional average $E[p | z]$ instead. The gain comes from denoising toward the right neighbourhood — which also explains why transport can create errors as well as fix them.

Prototypes are built once, in Stage 1, and never rebuilt:

$$\mu_c = \text{normalize}(\text{mean}_{i \in S_c} \text{normalize}(z_i))$$

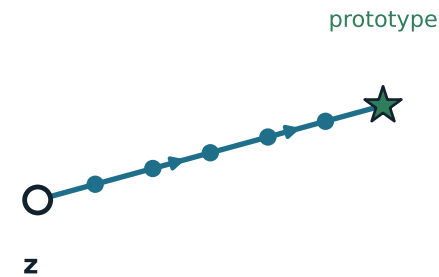
The velocity network, and the two ways to train it

$v(z, t)$ — 2 hidden layers, width 512, SiLU



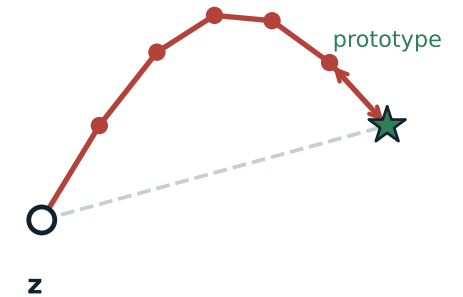
scalar t is concatenated to the feature, so one network covers the whole path. $D = 512 / 384 / 1024$

Standard FM



loss on the velocity at random t along the ideal straight path

Rolled-out FM



loss only on the endpoint, backprop through all T steps

$$\mathcal{L}_{FM} = \|v_{\theta}(z_t, t) - (p_y - z)\|^2$$

$$\mathcal{L}_{roll} = \|\hat{z}_T - p_y\|^2$$

Why this asymmetry matters

The standard objective never mentions T , so ONE network is trained per setting and evaluated at every T . The rolled-out objective bakes T in, so TWO networks are trained per setting, one per T , and a $T=4$ network is never run at $T=12$.

Getting this backwards either doubles standard FM's cost for nothing, or silently evaluates a model at a T it never saw.

What each objective buys

Standard FM supervises the velocity densely along the ideal path, but is trained on points it will never actually visit at inference. Rolled-out training removes that mismatch — it runs the real T -step sequence — but supervises only where the path ends, leaving the field free to become degenerate in between. The measured consequence is on page 7.

Experimental protocol

Stage 2 reuses Stage 1's splits, subsets, seeds, cached features and saved prototypes without recomputing any of them, so ΔAcc isolates the layer rather than a change in the data.

Datasets and official splits

dataset	classes	rule	train / val / test
DTD	47	partition 1	1,880 / 1,880 / 1,880
FGVC-Aircraft	100	variant level	3,334 / 3,333 / 3,333
Flowers-102	102	official	1,020 / 1,020 / 6,149

Training-set sizes and repetitions

setting	repeats	varied by
K = 5	3 runs	subset seeds 0,1,2
K = 10	3 runs	subset seeds 0,1,2
K = full	3 runs	init seeds 0,1,2

The grid

Per (dataset, encoder, K, seed): one standard network and two rolled-out networks (T = 4, 12).

image branch $3 \times 2 \times 3 \times 3 \times 3 = 162$ trainings
CLIP branch $3 \times 1 \times 3 \times 3 \times 3 = 108$ trainings

Test data is touched only after checkpoint selection.

Two reporting caveats we hold to

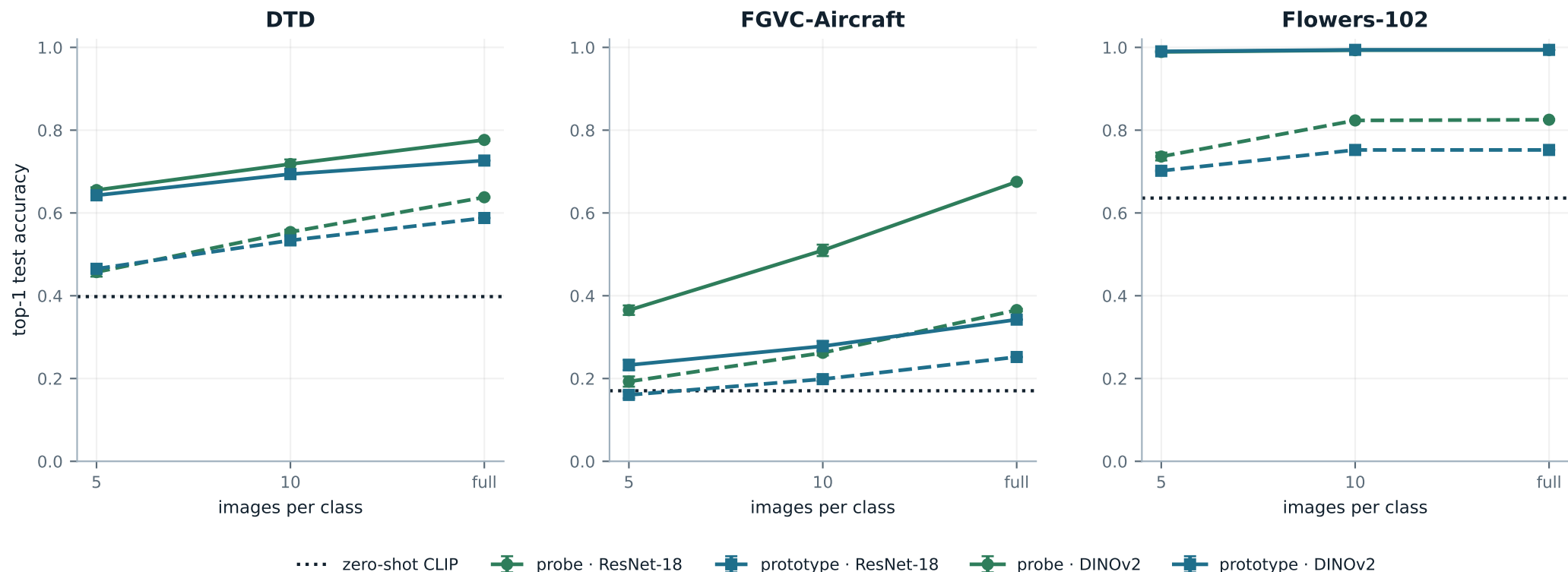
Flowers-102 at K = 10 has std = 0 by construction: the official train split is exactly 10 images per class, so all three subset seeds pick the same images. That is one effective run.

The Stage 1 image-prototype full setting is a single closed-form run, so it carries no error bar.

Held fixed between Stage 1 and Stage 2

- ✓ frozen encoders — never re-run, never fine-tuned
- ✓ cached features — the identical tensors Stage 1 used
- ✓ class prototypes — loaded from Stage 1, never rebuilt
- ✓ K-shot subsets — re-derived with the same seeded selection
- ✓ classification rule — the same cosine comparison

Stage 1 — three baselines on frozen features



What Stage 1 established

Encoder choice dominates every method difference in this project. DINOv2 beats ResNet-18 by 14 to 31 points on every dataset and every K — larger than any gap between the three baselines. The linear probe is strongest at full data everywhere except saturated Flowers-102/DINOv2; at K = 5 the closed-form prototype is competitive and sometimes wins.

Zero-shot CLIP RN50 (prescribed prompt)

DTD 0.398 Aircraft 0.170 Flowers-102 0.636
A 5-prompt ensemble adds +2.3 / -0.1 / +2.0 points.

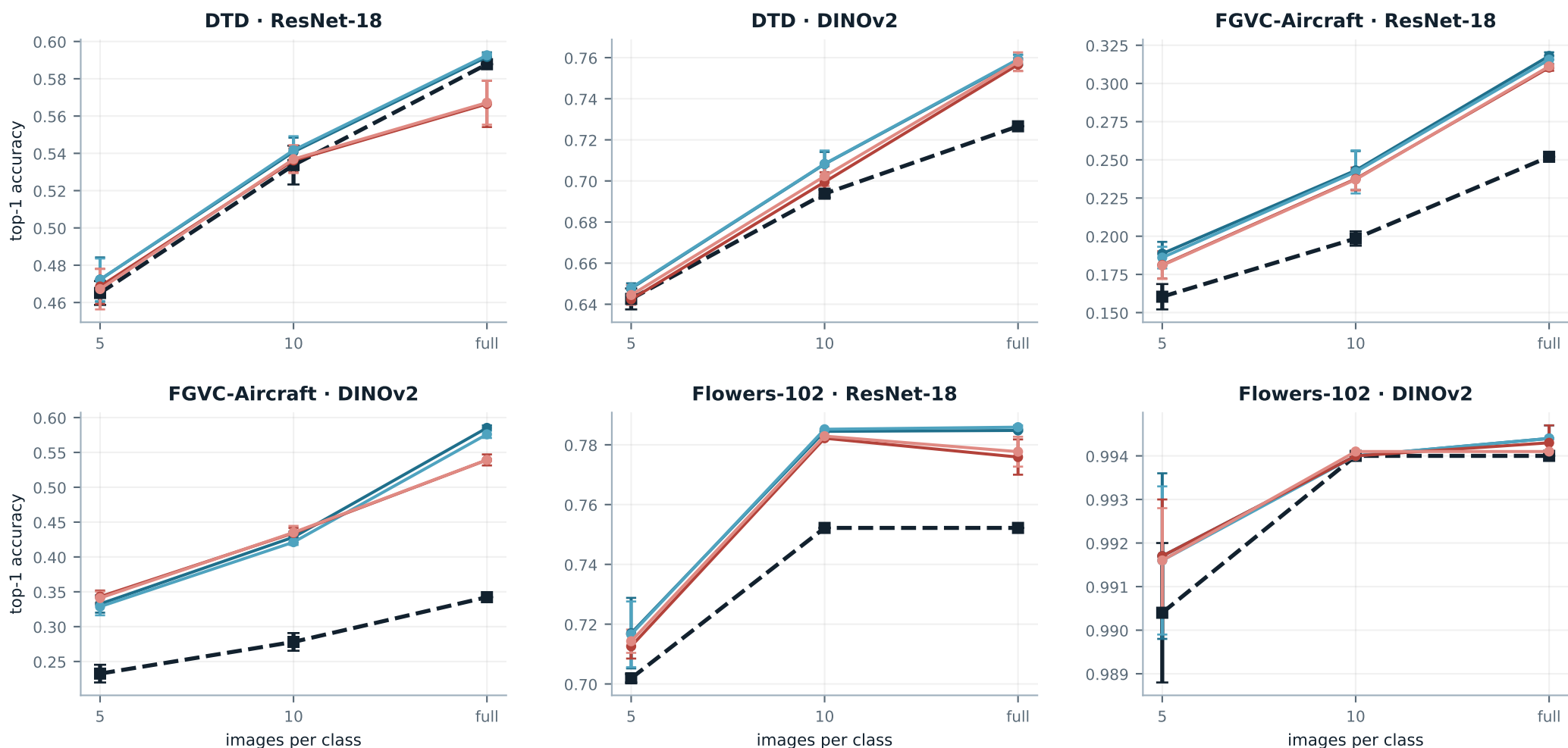
The one that surprises people

On Aircraft, zero-shot CLIP (.1704) beats 5-shot ResNet-18 prototypes (.1604). With 100 fine-grained variants, five labelled images per class buy less than a well-chosen text prompt.

Stage 2 — the FM layer versus the prototype baseline

Six panels, one per dataset × encoder. Black = Stage 1 baseline; blue = standard FM; red = rolled-out FM.

■ Stage 1 baseline
● standard T=4
● standard T=12
● rollout T=4
● rollout T=12



What the image branch shows

Best FM variant per cell, with ΔAcc against the matching Stage 1 baseline

dataset · encoder	K = 5	K = 10	K = full
DTD · ResNet-18	0.4723 (+0.0072)	0.5417 (+0.0080)	0.5926 (+0.0048)
DTD · DINOv2	0.6480 (+0.0054)	0.7083 (+0.0145)	0.7592 (+0.0326)
FGVC-Aircraft · ResNet-18	0.1886 (+0.0282)	0.2431 (+0.0446)	0.3180 (+0.0660)
FGVC-Aircraft · DINOv2	0.3430 (+0.1104)	0.4351 (+0.1570)	0.5852 (+0.2429)
Flowers-102 · ResNet-18	0.7170 (+0.0151)	0.7852 (+0.0330)	0.7859 (+0.0337)
Flowers-102 · DINOv2	0.9917 (+0.0013)	0.9941 (+0.0001)	0.9944 (+0.0004)

Across all 72 result cells

66 improved

3 flat (± 0.0000)

3 regressed

All three regressions are rolled-out training.

Four findings

1 The gain grows with K, and is largest where the baseline is weakest.

Aircraft/DINOv2 goes +11.0 $\rightarrow$ +15.7 $\rightarrow$ +24.3 points. This is the signature of a trained layer: a class mean stops absorbing information, while the velocity field keeps using the extra labels.

2 It closes much of the prototype-to-probe gap, but never passes the probe.

At full data FM recovers roughly 20–75% of the distance to the linear probe and overtakes it only on saturated Flowers-102/DINOv2, where the difference is inside seed noise.

3 Standard FM beats rolled-out training in 14 of 18 settings (rolled-out wins 3, 1 tie).

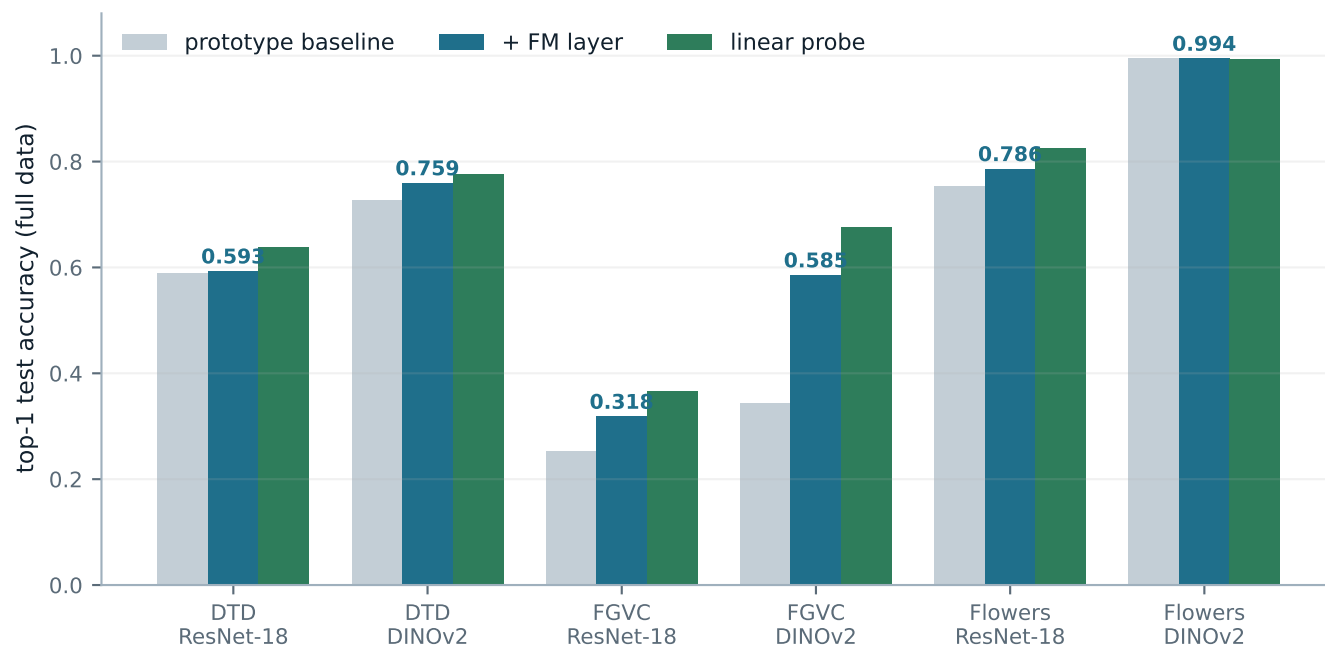
Rolled-out is also the only variant that ever regresses below the baseline, and costs 1.5–2.5 $\times$ more per run. Endpoint-only supervision constrains the field far more weakly than per-step velocity regression.

4 T is nearly irrelevant — and this is expected, not a bug.

The ideal velocity $p - z$ is constant along the path, so a well-learned field lands in the same place for any T. An identical-looking T=4 / T=12 column otherwise reads as a copy-paste error, so say it before being asked.

The comparison that matters most

At full data: the prototype baseline, the same baseline with an FM layer, and the Stage 1 linear probe on identical features.



Read this honestly

The FM layer is a real improvement over the prototype rule it replaces — clearly so, and by a lot on Aircraft/DINOv2 (.342 → .585).

It is not a replacement for a discriminatively trained classifier. The probe leads in five of six settings, and the one FM win is on a task already at 99.4%.

Also worth stating: ΔAcc bundles "added an FM layer" with "added a trained, validation-selected ~0.8M-parameter head", against a baseline with zero trained parameters and no access to the validation split.

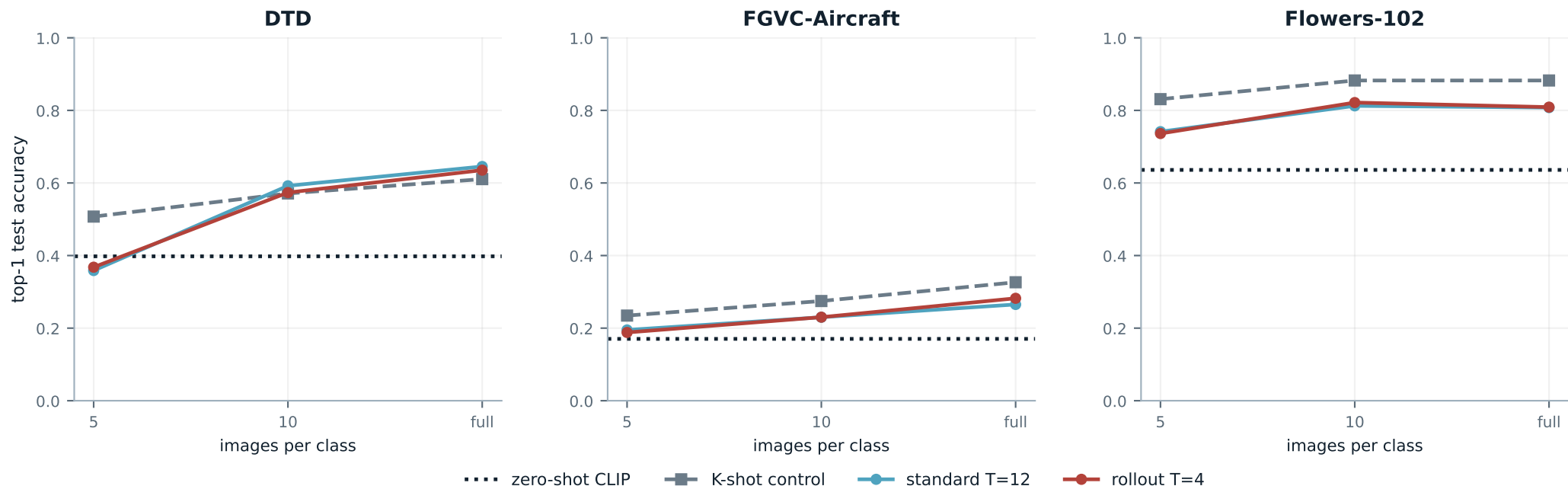
Training behaviour

Both objectives trained stably; nothing diverged across 162 runs. 161 of 162 early-stopped and one reached the 200-epoch cap. In 16 runs (~10%) the best validation accuracy was at epoch 1 — concentrated in the saturated Flowers-102/DINOv2 settings, where there is nothing left to gain. Those FM layers are effectively untrained and should not be described otherwise.

The per-epoch loss curves for both objectives are in the notebook (Section 9); they are omitted here because at page scale the six-panel grid is not legible, and the claim it supports - stable training, no divergence - is one sentence.

The CLIP branch — why the control changes the story

Here FM transports CLIP image embeddings toward frozen text prototypes. Because the layer trains on labels, this is no longer zero-shot — so a same-supervision control is required: image prototypes built from the same CLIP features and the same K-shot subsets.



Against zero-shot: large gains

FM beats the zero-shot baseline in 32 of 36 cells, by as much as +24.8 points on DTD at full data.

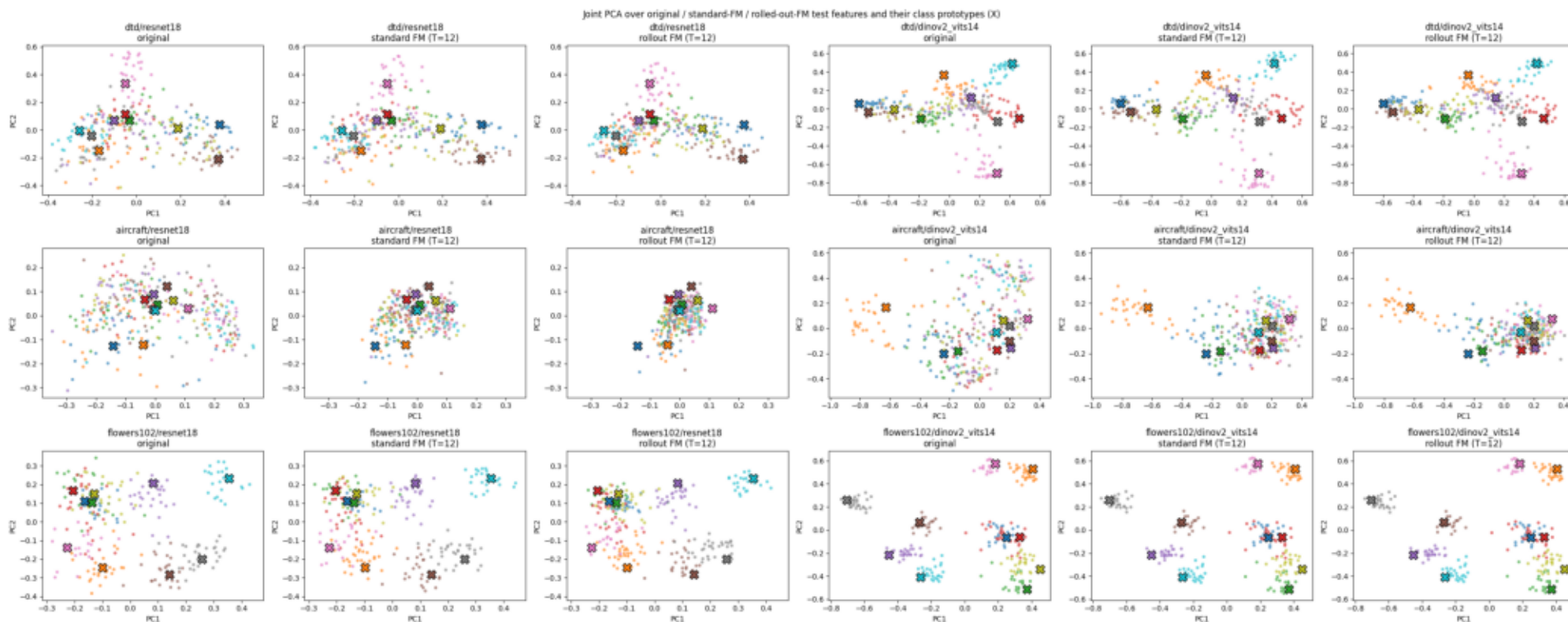
Taken alone this looks like a decisive win for the FM layer. It is not — the FM side used K labelled images per class and the zero-shot side used none.

Against the fair control: it mostly loses

Given the same labels, simply building image prototypes beats transporting toward text prototypes in 28 of 36 cells. Only DTD at K = 10 and K = full come out ahead (2 of 9 settings).

On DTD at K = 5 the FM layer is even worse than zero-shot itself.

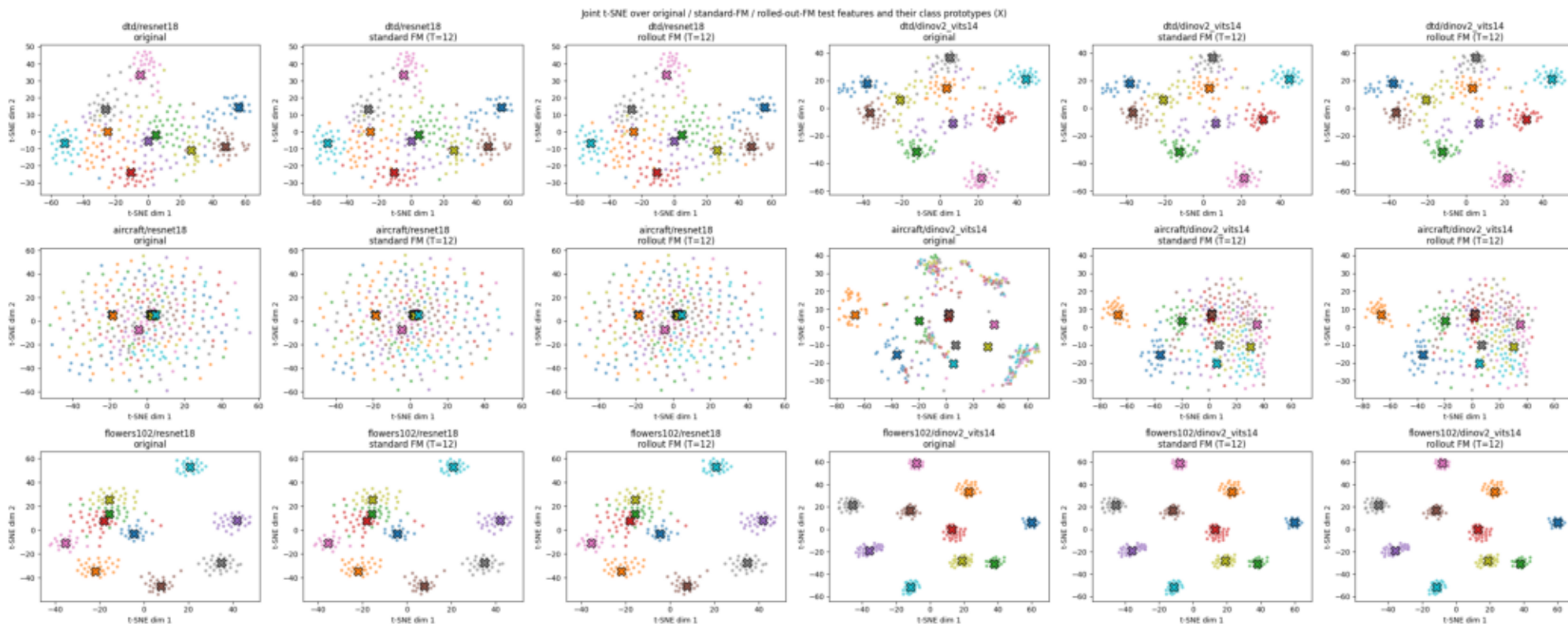
Feature-space geometry — what the layer actually does



Test features before the flow, after standard FM, and after rolled-out FM, with the class prototypes (X). One PCA is fitted jointly across all three views and the prototypes, so a point's movement between panels is a real displacement in one shared plane.

The mechanism is visible directly: overlapping class clouds are contracted onto their prototypes. Because the network cannot see the label at test time, points in ambiguous regions are pulled toward whichever prototype's basin they fall in — which is why transport can create errors as well as fix them, and why ΔAcc can be negative even when the training loss is low.

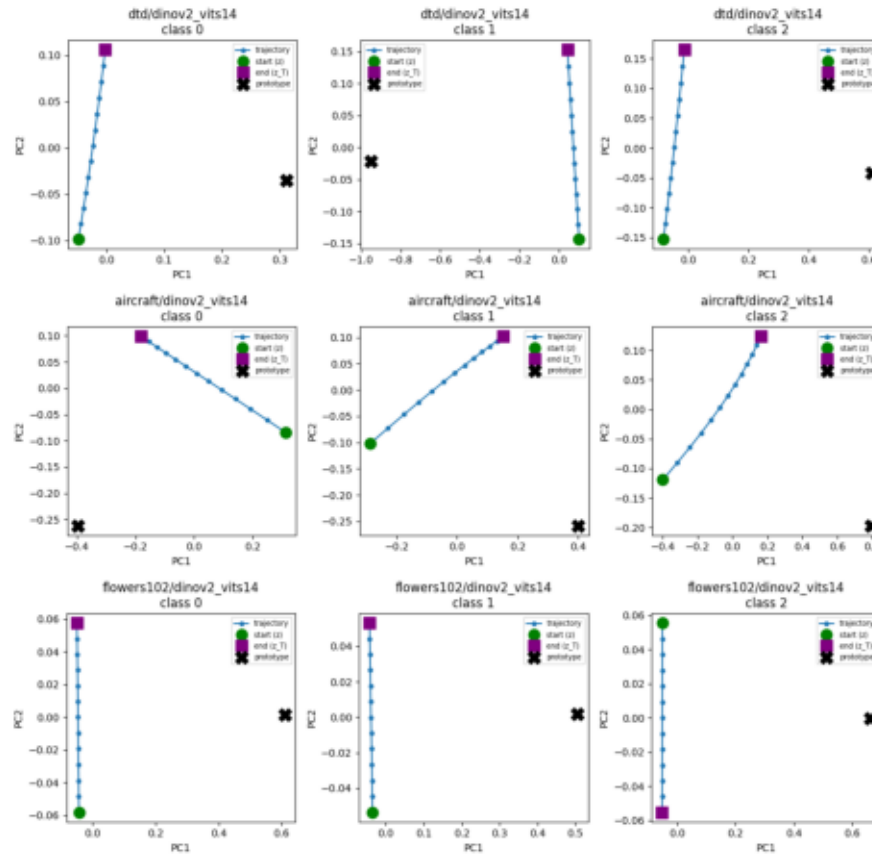
The same comparison under a joint t-SNE



The identical transported tensors under t-SNE instead of PCA, fitted once per row over all three views and the prototypes. Cluster separation reads well here; distances do not.

Read this qualitatively and use the PCA page for anything geometric. t-SNE normalises local density, so it deliberately re-expands the tightly contracted post-FM clusters — the visual shrinkage looks far milder here than it is numerically.

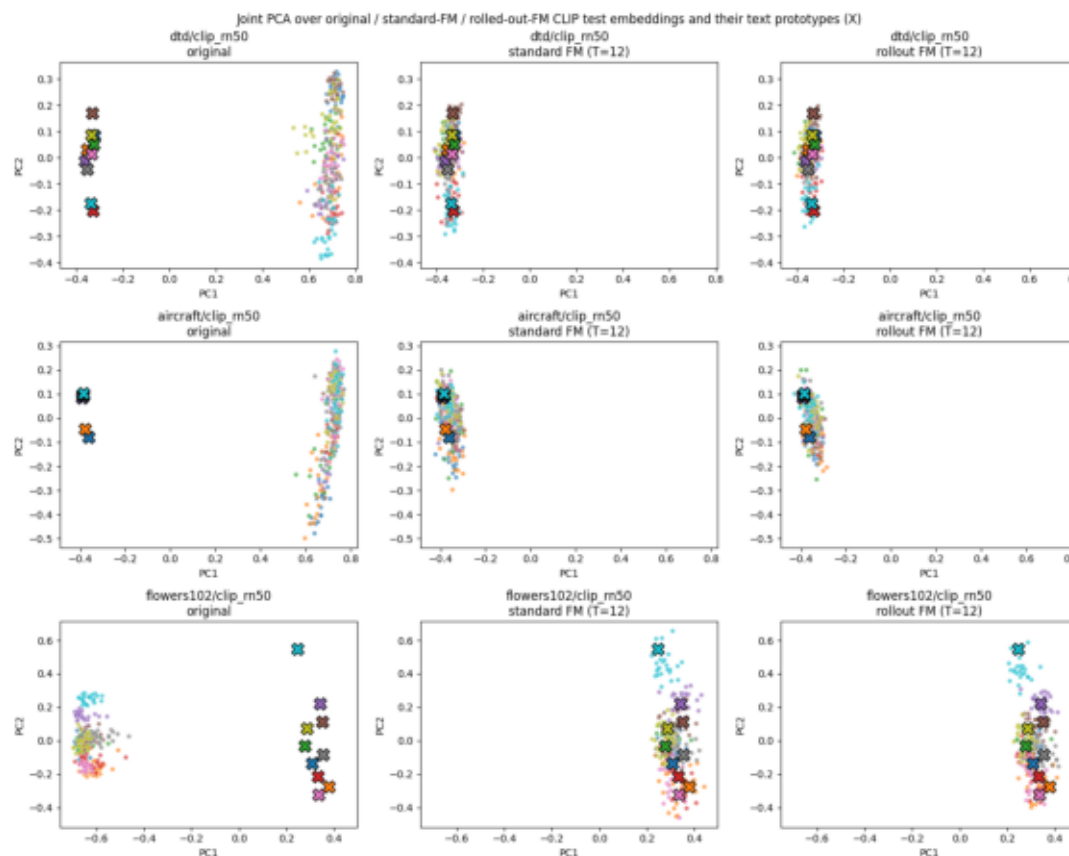
Flow trajectories — the transport step by step



Intermediate Euler states for representative test examples, from the original feature (green) through every step to the transported endpoint (purple square) and the target prototype (X). PCA rather than t-SNE, so a straight path stays a straight path.

The paths are close to straight, which is the geometric counterpart of the T-independence result: when the learned field matches the ideal constant velocity, four steps and twelve steps arrive at the same place.

The CLIP modality gap, and the flow across it

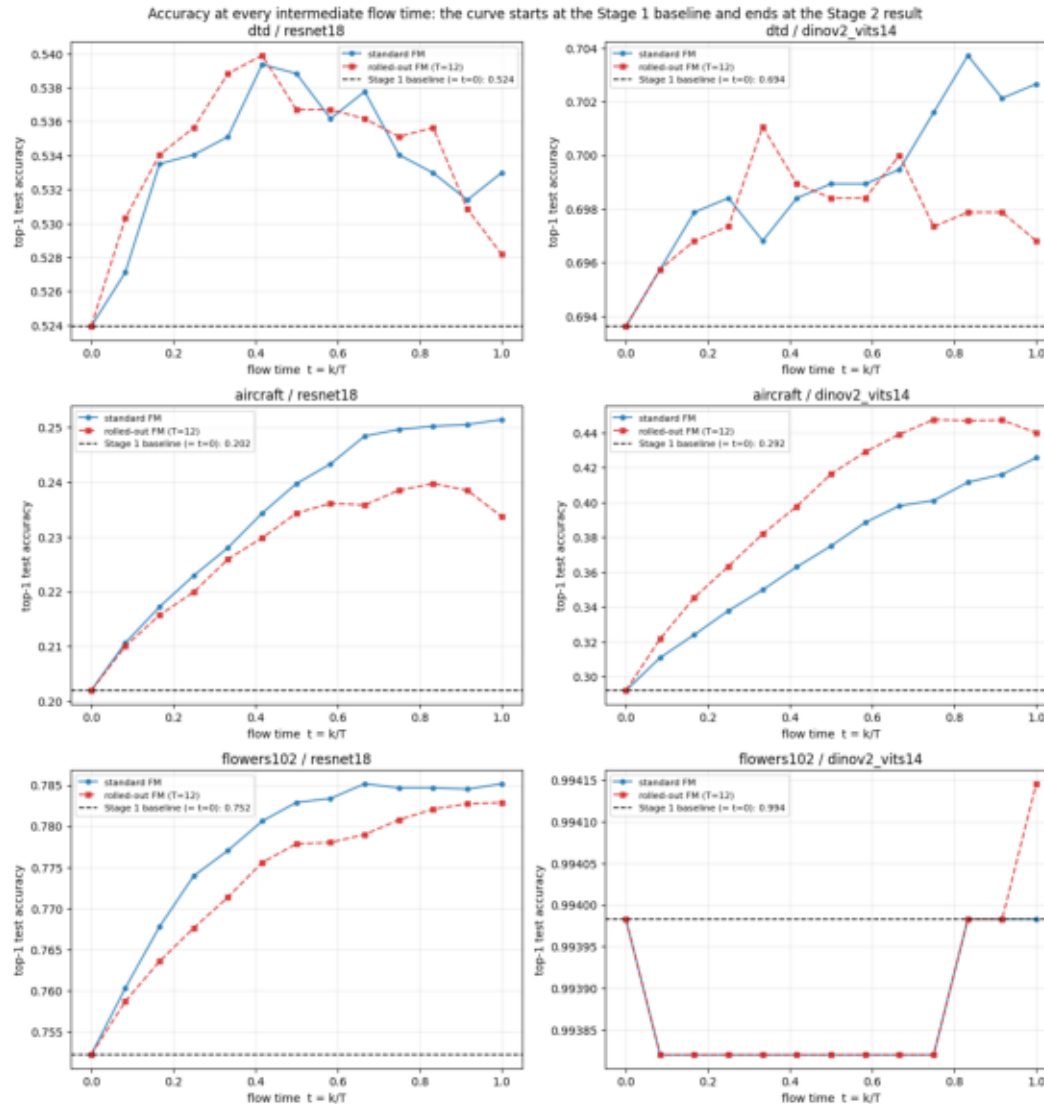


The same three-view comparison on the CLIP branch. The image embeddings and the text prototypes (X) start in visibly different regions — that separation is CLIP's image-text modality gap — and the learned transport carries the image cloud across it.

This is the qualitative reason the CLIP branch behaves differently from the image branch: the transport has to cross a systematic offset between two modalities, not merely denoise within one feature space.

Samples and prototypes at intermediate flow times

Rather than looking only at $t=0$ and $t=1$, every intermediate Euler state is classified over the complete test split. The $t=0$ end is asserted, not assumed - and both assertions passed, so each curve provably starts at the baseline and ends at the FM result.



Δ Acc, unrolled

Each curve begins exactly on the dashed Stage 1 baseline and ends exactly at the Stage 2 number. The shape between them is what the layer does.

Two things are visible immediately:

Aircraft climbs steeply and keeps climbing: +5.0 and +13.4 points in this 10-shot setting, and up to +24.3 at full data.

DTD and Flowers are nearly flat. The transport is running, the features are moving, and the classification barely changes.

The next page explains why, using the margin.

Why the gains land where they do

Two quantities separate "the flow moved the point" from "the flow classified it better": the cosine to the true prototype, and the margin — cosine to the true prototype minus the best competing one. The first says transport happened; only the second says it helped.

setting (10-shot, seed 0, standard FM)	cosine to true prototype	margin	accuracy	Δ
DTD · ResNet-18	0.726 → 0.819	+0.0051 → +0.0035	0.5239 → 0.5330	+0.0091
DTD · DINOv2	0.544 → 0.639	+0.1135 → +0.1110	0.6936 → 0.7027	+0.0091
Flowers-102 · DINOv2	0.815 → 0.821	+0.3142 → +0.3061	0.9940 → 0.9940	+0.0000
FGVC-Aircraft · ResNet-18	0.889 → 0.942	-0.0160 → -0.0072	0.2019 → 0.2514	+0.0495
FGVC-Aircraft · DINOv2	0.709 → 0.883	-0.0542 → -0.0088	0.2919 → 0.4257	+0.1338

The rule

Margin rises (green) only where the baseline was already mis-ranked — negative margin, i.e. the true prototype was not the top match.

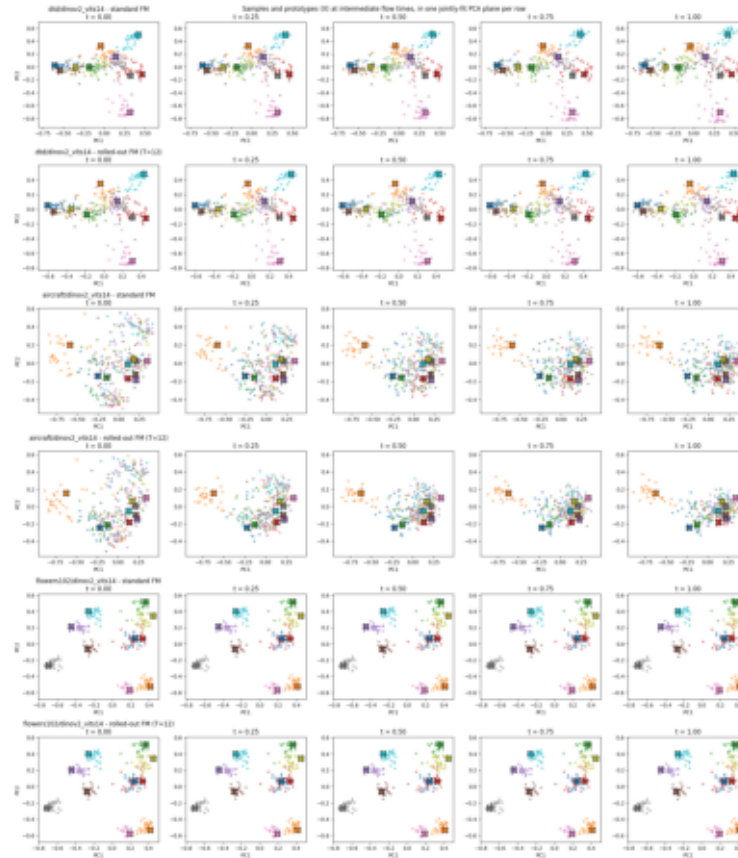
Those are exactly the settings where accuracy jumps.

The FM layer repairs a broken classifier; it does not sharpen a working one.

Cosine to the true prototype rises in every single setting — the transport always does what it was trained to do. But where the prototype baseline already ranked classes correctly (DTD, Flowers-102: positive margin), pulling every point toward every prototype shrinks the margin slightly, and accuracy moves by less than a point. Where the baseline was mis-ranked (Aircraft: negative margin), the flow repairs the ranking — +.045 of margin on Aircraft/DINOv2 — and accuracy jumps 13.4 points.

This is the measured mechanism behind the headline finding that gains are largest where the baseline is weakest. It also predicts the ceiling: once a margin is comfortably positive, prototype-targeted transport has nothing left to fix.

The point cloud at intermediate flow times



Samples and prototypes (X) at $t = 0, 1/4, 1/2, 3/4, 1$. Each row is one dataset at its strongest encoder, once per training objective; one PCA is fitted jointly over every snapshot plus the prototypes, and all five panels of a row share axis limits.

The contraction is the mechanism made visible: overlapping class clouds are drawn onto their prototypes. Standard FM moves points along nearly straight, roughly parallel paths; rolled-out FM, supervised only at the endpoint, takes a more curved route and contracts later.

$t = 1$ is a convention, not an optimum

Accuracy peaks before the endpoint in 6 of 6 CLIP-branch curves and 6 of 12 image-branch curves. Integrating the full T steps therefore costs accuracy in most settings — and the loss is largest for rolled-out models on the CLIP branch.

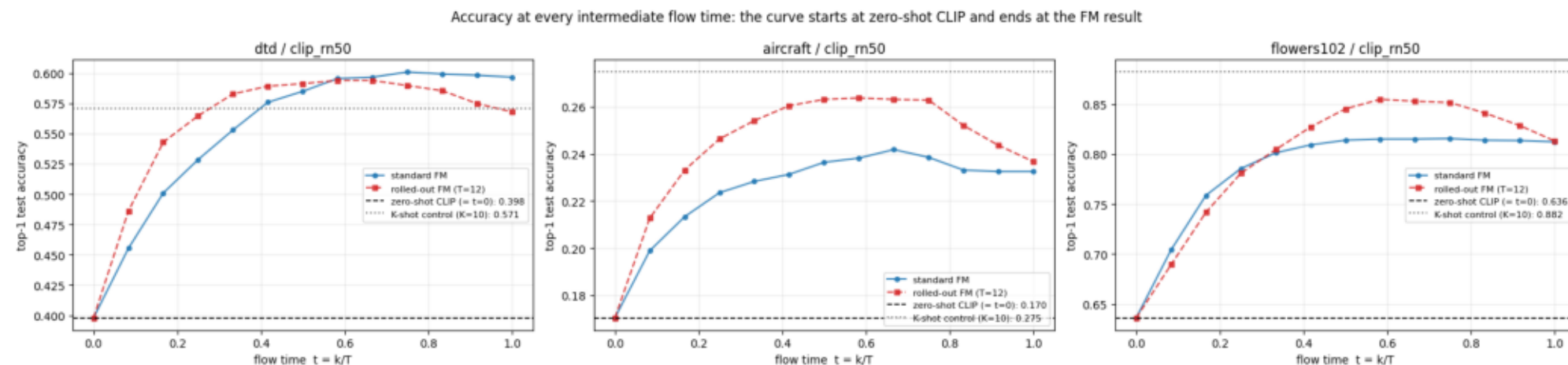
setting	objective	best t	acc at best t	acc at $t=1$	left on table
CLIP · Flowers-102	rolled-out	0.58	0.8551	0.8133	+4.18 pts
CLIP · FGVC-Aircraft	rolled-out	0.58	0.2637	0.2367	+2.70 pts
CLIP · DTD	rolled-out	0.58	0.5941	0.5681	+2.60 pts
image · DTD / ResNet-18	rolled-out	0.42	0.5399	0.5282	+1.17 pts
CLIP · FGVC-Aircraft	standard	0.67	0.2418	0.2325	+0.93 pts
image · FGVC-Aircraft / DINOv2	rolled-out	0.75	0.4476	0.4401	+0.75 pts
image · DTD / ResNet-18	standard	0.42	0.5394	0.5330	+0.64 pts
image · FGVC-Aircraft / ResNet-18	rolled-out	0.83	0.2397	0.2337	+0.60 pts

What to do with this

The stopping time is free to choose and can be selected on validation data like any other hyperparameter.

On the CLIP branch it is worth more than the choice between the two training objectives.

Rolled-out models overshoot hardest, which fits their weaker constraint: only the endpoint is supervised, so nothing discourages the field from running past the useful region.



CLIP branch: every curve starts on the dotted zero-shot line, and every one peaks before $t = 1$.

The flow in reverse — and what it reveals about CLIP

Start from the class prototypes at $t = 1$ and integrate the same learned field backward. The recovery rate asks whether a prototype, after that round trip, lands nearest to its own class's mean test feature. The untouched forward prototype is the control and ceiling.

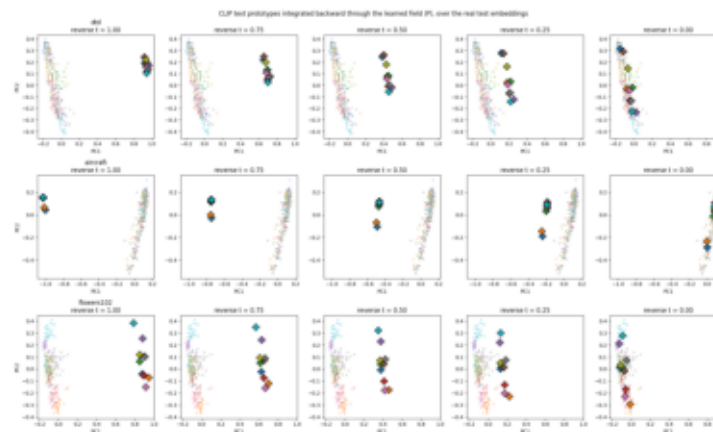
branch · setting	forward	reverse	cosine distance moved
image · DTD / ResNet-18	0.957	0.957	0.033
image · DTD / DINOv2	1.000	1.000	0.026
image · FGVC-Aircraft / ResNet-18	0.660	0.530	0.026
image · FGVC-Aircraft / DINOv2	0.690	0.620	0.044
image · Flowers-102 / ResNet-18	1.000	1.000	0.014
image · Flowers-102 / DINOv2	1.000	1.000	0.005
CLIP · DTD	0.553	1.000	0.718
CLIP · FGVC-Aircraft	0.250	0.750	0.697
CLIP · Flowers-102	0.745	1.000	0.677

Image branch: almost nothing moves

Cosine distances of .006 to .044. The learned field is essentially zero near $t = 1$, so backward integration does not travel back toward the feature distribution. Recovery is preserved except on Aircraft, where it degrades. Well-behaved, but not usefully invertible — which is what a contraction should look like.

CLIP branch: the modality gap, measured

The prototypes move nearly orthogonally ($\approx .70$) and recovery jumps from .55 / .25 / .75 to 1.00 / .75 / 1.00. The forward rate measures how far CLIP's text embeddings sit from its image embeddings; the reverse pass closes that distance. Direct evidence that the field learned to bridge the gap.

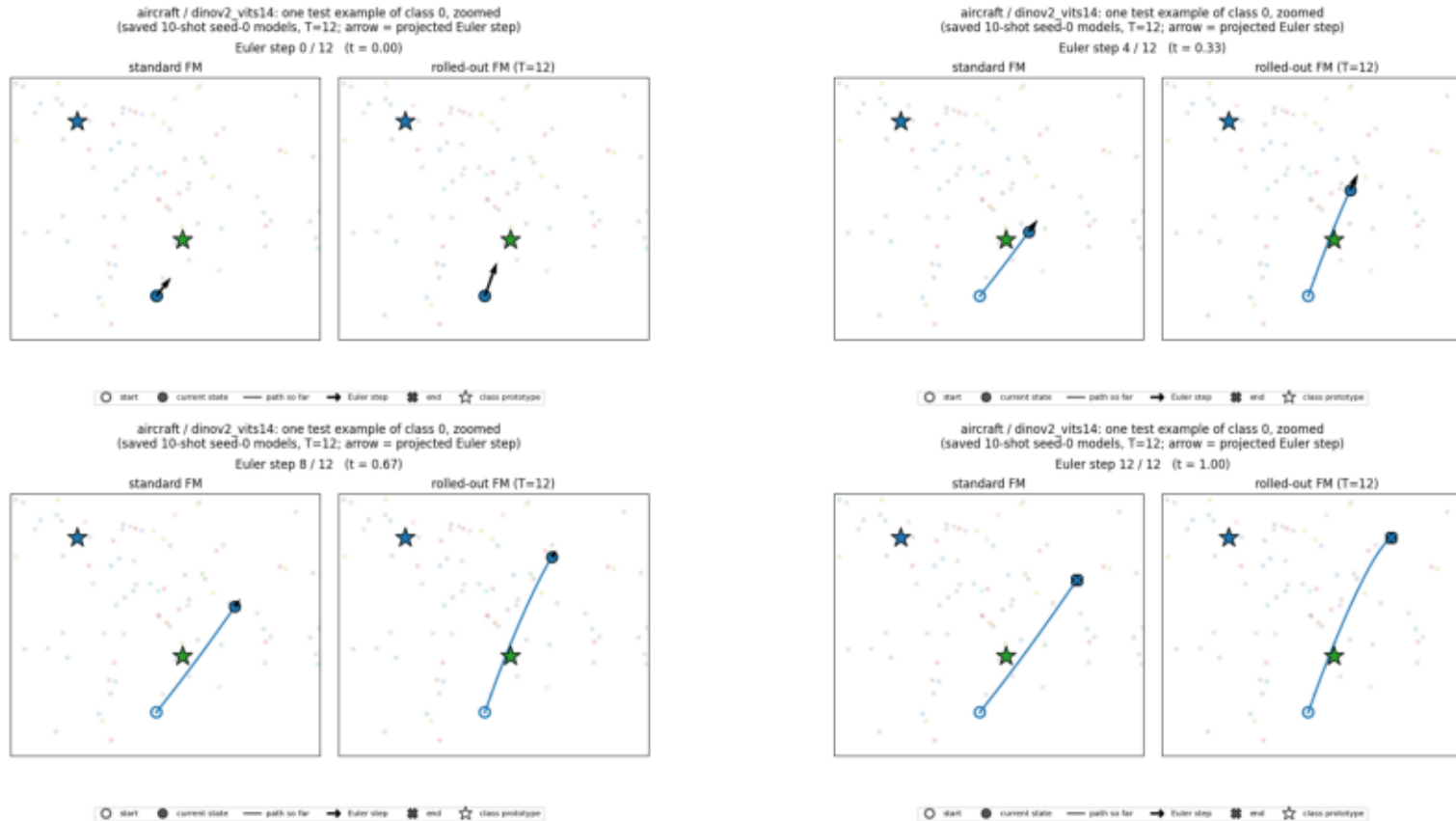


CLIP text prototypes (P) integrated backward over the real test embeddings, at reverse $t = 1 \rightarrow 0$. They start off to one side — that is the gap — and move in.

Flow Matching as a classification layer · Stages 1-2

The transport, frame by frame

Four frames from the animation the notebook exports as a GIF: standard FM (left panel) and rolled-out FM (right) starting from an identical test feature, with the arrow showing the actual Euler update at that step. The setting is chosen programmatically as the largest-gain one — Aircraft / DINOv2, where the layer is worth +24.3 points.



Read qualitatively: the transport happens in DINOv2's full 384-dimensional space and this is a 2-D projection of it, so projected path length and curvature are not high-dimensional measurements. What the frames convey is the mechanism — T small updates accumulating into the representation the cosine rule finally classifies.

Findings

- 1 **Encoder choice dominates every method difference measured here.**
DINOv2 over ResNet-18 is worth 14-31 points. No baseline-versus-baseline gap in this project comes close.
- 2 **The FM layer works: it improves the prototype baseline in 66 of 72 cells.**
The gain grows with K and is largest where the baseline is weakest - up to +24.3 points on Aircraft/DINOv2 at full data.
- 3 **It repairs a broken classifier rather than sharpening a working one.**
Measured at intermediate flow times: cosine to the true prototype rises everywhere, but margin rises only where the baseline was mis-ranked.
- 4 **It does not beat a linear probe on the same features.**
It recovers 20-75% of the prototype-to-probe gap and only overtakes where accuracy is already saturated.
- 5 **Standard FM $\geq$ rolled-out training (14/18 settings), on accuracy and on compute.**
Rolled-out is the only variant that ever regresses below the baseline, and the one that overshoots hardest.
- 6 **$t = 1$ is a convention, not an optimum.**
Accuracy peaks earlier in 6/6 CLIP curves and 6/12 image curves - up to +4.2 points left on the table.
- 7 **On the CLIP branch, the fair control reverses the headline.**
FM beats zero-shot in 32/36 cells but loses to a same-supervision control in 28/36. Those gains measure the labels.
- 8 **The reverse flow makes CLIP's modality gap measurable.**
Backward integration lifts text-prototype recovery from .55/.25/.75 to 1.00/.75/1.00; on the image branch it barely moves anything.

For Stage 3

Stage 3 places the FM module before a linear classifier. Two Stage 2 lessons carry over: the transport map has real discriminative capacity, but prototype-targeted transport alone does not recover a trained decision boundary — and per-path supervision is the stable optimisation regime, while endpoint-only objectives are the ones that regressed here.

Limitations and next steps

Limitations

- **n = 3 runs per setting.**
Spreads are sample standard deviations; no confidence intervals are quoted at this sample size.
- **Flowers-102 at K = 10 is one effective run.**
The official train split is exactly 10 images per class, so all three subset seeds select the same images; its std is 0 by construction.
- **The Stage 1 full prototype baseline is a single run.**
It is closed-form, so it carries no error bar and the full-data deltas are unpaired.
- **No hyperparameter search.**
The Stage 1 probe recipe was adopted as-is, per the specification. Better FM numbers are plausibly reachable with validation-only tuning.
- **2-D projections are qualitative.**
The joint PCA plane explains only part of the variance; geometry off-plane is invisible. They are not evidence of classifier quality.
- **The CLIP branch cannot isolate the FM effect against zero-shot.**
That is exactly why the same-supervision control is reported alongside it.

Next steps, in order of expected value

Both optional extensions are now implemented and executed in both notebooks; the pages above report their measured results. What they surfaced points at three concrete follow-ups:

- > **Make the stopping time a hyperparameter.**
Accuracy peaks before $t = 1$ in every CLIP-branch setting and in most image-branch ones, worth up to +4.2 points. It is selectable on validation data at no cost, and on the CLIP branch it matters more than the choice between the two training objectives.
- > **Use the margin, not accuracy, to predict where the layer will help.**
A negative baseline margin marks a mis-ranked classifier, which is precisely where prototype-targeted transport pays. A comfortably positive margin predicts the near-zero gains seen on DTD and Flowers-102.
- > **Carry the reverse-flow finding into any CLIP work.**
The forward recovery rate is a cheap, direct measure of the modality gap, and the reverse pass shows a learned field can close it.